How to Install Docker on Raspberry Pi

December 12, 2019

DEBIAN DOCKER RASPBIAN

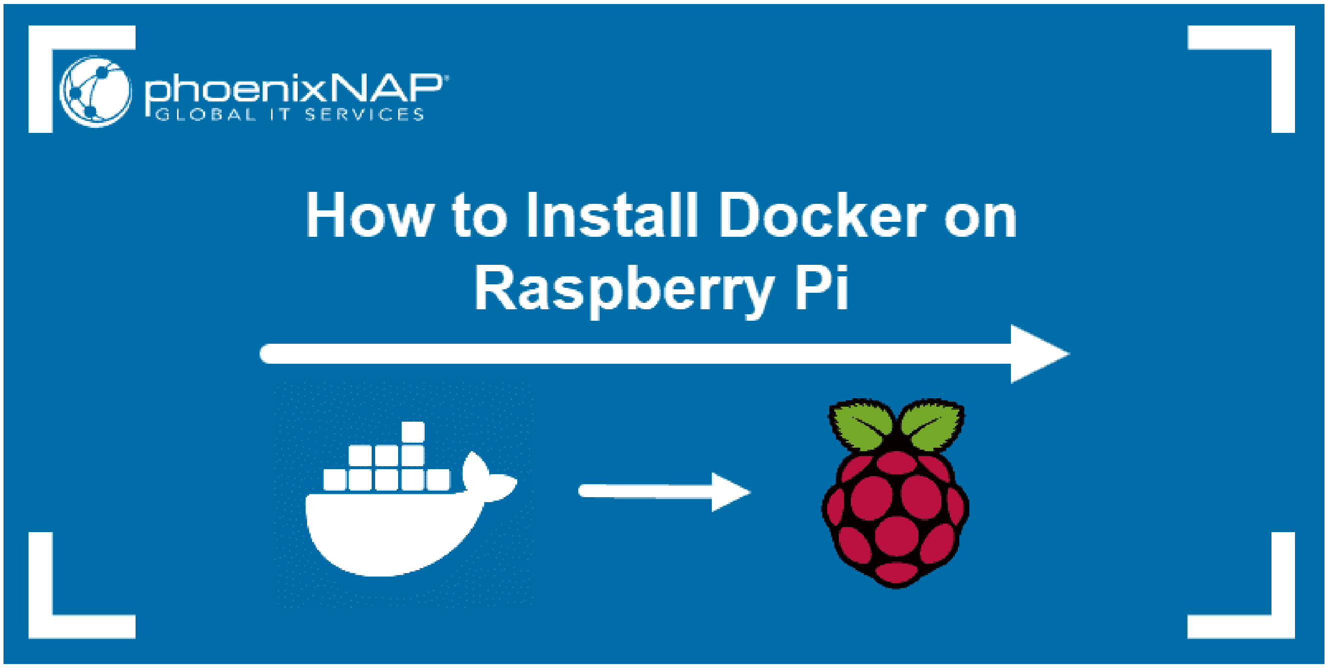
Home » SysAdmin » How to Install Docker on Raspberry Pi

Introduction

Docker is a tool for creating, deploying, and running applications in containers. The software is popular among developers as it speeds up the development process and does not use a lot of resources.

Docker containers are lightweight, especially compared to virtual machines. This feature is especially valuable if you are a Raspberry Pi user.

If you need help installing Docker on your Raspberry Pi, read our step-by-step guide on **how to install Docker on Raspberry Pi**.



Prerequisites

- Raspberry Pi with a running Raspbian OS
- Raspbian Stretch (Lite)
- SSH connection enabled

How to Install Docker on Raspberry Pi

To install Docker on your Raspberry Pi, you need to go through the following steps:

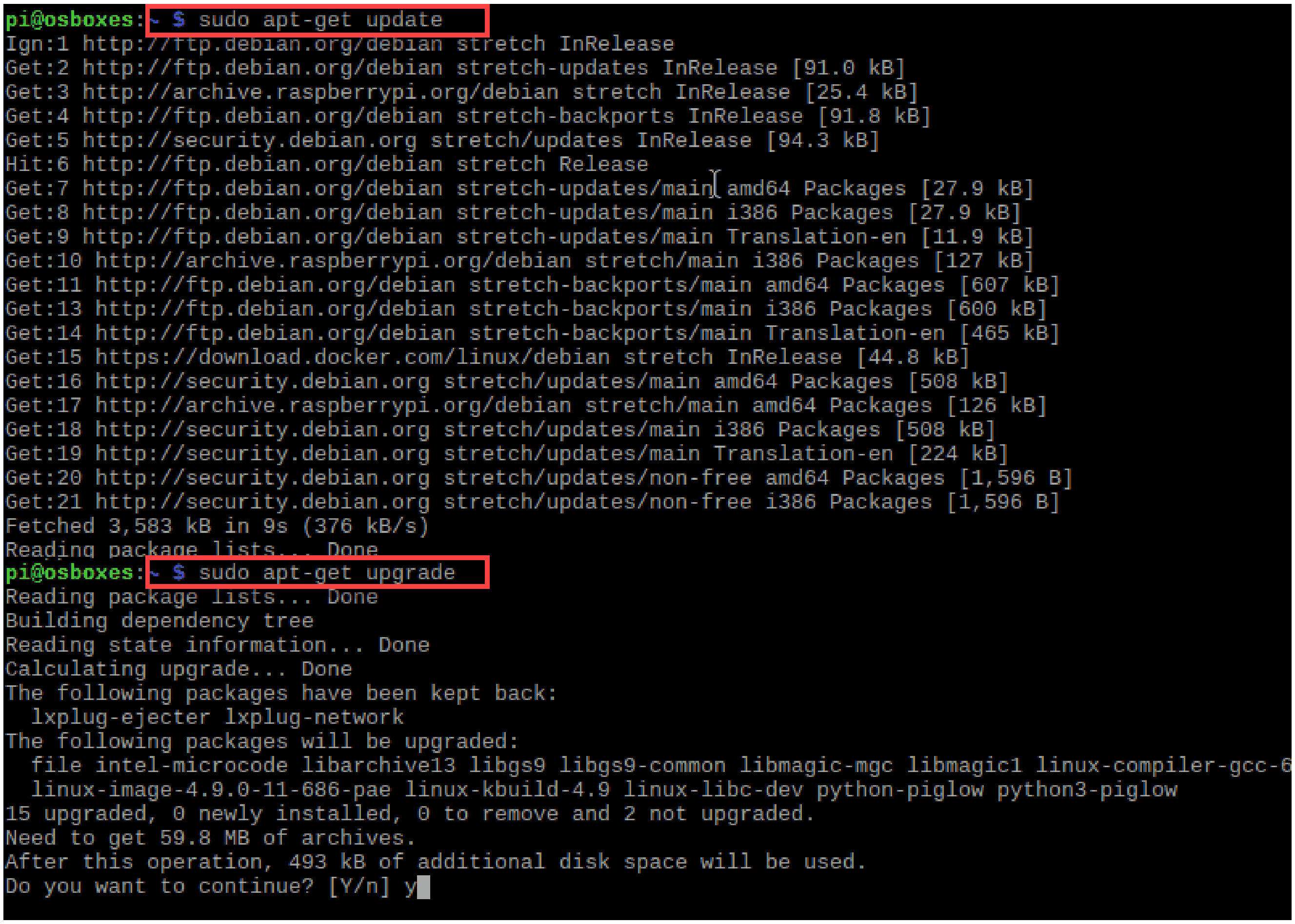
1. Update and upgrade your system.
2. Download the installation script and install the package.
3. Allow a non-root user to execute Docker commands.
4. Verify installation by checking the Docker version.
5. Test the set up by running a "hello-world" container.

Step 1: Update and Upgrade

Start by updating and upgrading the system. This ensures you install the latest version of the software.

Open a terminal window and run the command:

```
sudo apt-get update && sudo apt-get upgrade
```



Note: To avoid security and performance issues make sure to update Raspberry Pi frequently.

Step 2: Download the Convenience Script and Install Docker on Raspberry Pi

Move on to downloading the installation script with:

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

Execute the script using the command:

```
sudo sh get-docker.sh
```

This installs the required packages for your Raspbian Linux distribution.

```
pi@osboxes:~ $ curl -fsSL https://get.docker.com -o get-docker.sh
pi@osboxes:~ $ sudo sh get-docker.sh
# Executing docker install script, commit: f45d7c11389849ff46a6b4d94e0dd1ffebca32c1
+ sh -c apt-get update -qq >/dev/null
```

The output will tell you which version of Docker is now running on your system.

```
Server: Docker Engine - Community
 Engine:
  Version:          19.03.5
  API version:      1.40 (minimum version 1.12)
  Go version:       go1.12.12
  Git commit:       633a0ea838
  Built:            Wed Nov 13 07:28:22 2019
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          1.2.10
  GitCommit:        b34a5c8af56e510852c35414db4c1f4fa6172339
 runc:
  Version:          1.0.0-rc8+dev
  GitCommit:        3e425f80a8c931f88e6d94a8c831b9d5aa481657
 docker-init:
  Version:          0.18.0
  GitCommit:        fec3683
If you would like to use Docker as a non-root user, you should now consider
adding your user to the "docker" group with something like:

  sudo usermod -aG docker your-user

Remember that you will have to log out and back in for this to take effect!

WARNING: Adding a user to the "docker" group will grant the ability to run
         containers which can be used to obtain root privileges on the
         docker host.
         Refer to https://docs.docker.com/engine/security/security/#docker-daemon-attac
k-surface
         for more information.
```

Step 3: Add a Non-Root User to the Docker Group

By default, only users who have administrative privileges (root users) can run containers. If you are not logged in as the root, one option is to use the **sudo** prefix.

However, you could also add your non-root user to the Docker group which will allow it to execute docker commands.

The syntax for adding users to the Docker group is:

```
sudo usermod -aG docker [user_name]
```

To add the **Pi** user (the default user in Raspbian), use the command:

```
sudo usermod -aG docker Pi
```

There is no specific output if the process is successful. For the changes to take place, you need to log out and then back in.

Step 4: Check Docker Version and Info

Check the version of Docker on your Raspberry Pi by typing:

```
docker version
```

The output will display the Docker version along with some additional information.
For system-wide information (including the kernel version, number of containers and images, and more extended description) run:

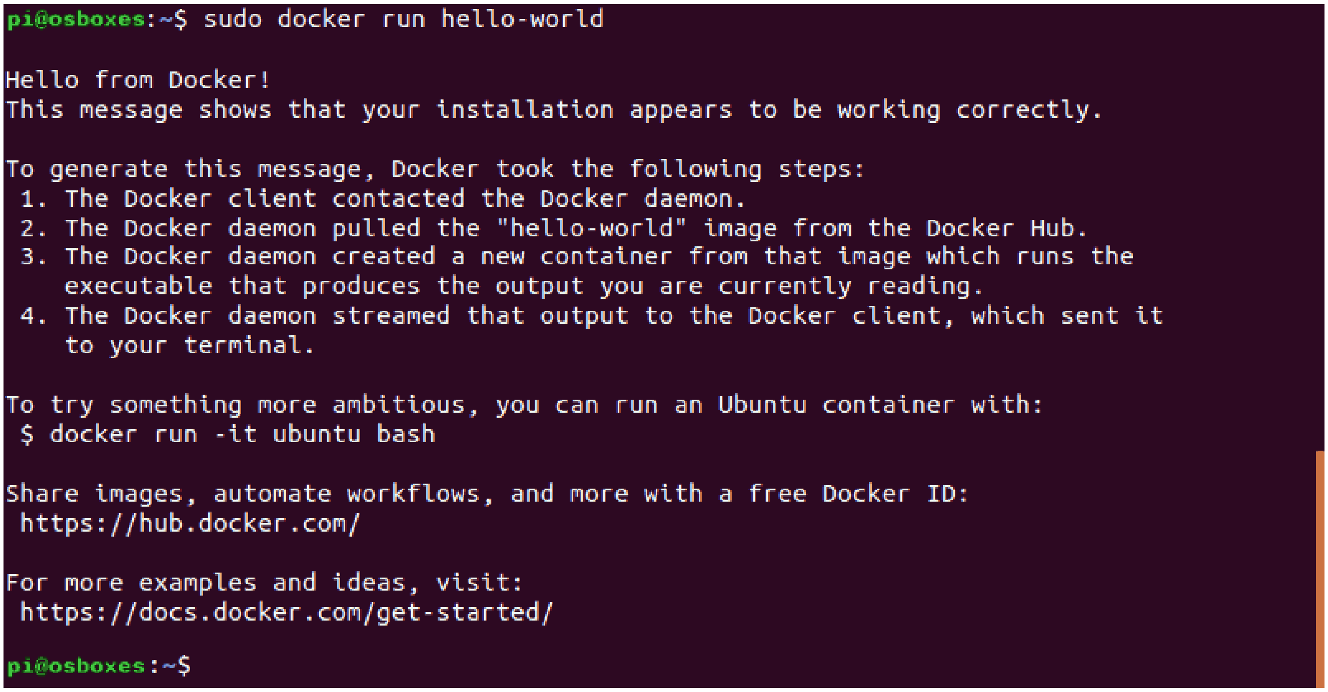
```
docker info
```

Step 5: Run Hello World Container

The best way to test whether Docker has been set up correctly is to run the *Hello World* container.
To do so, type in the following command:

```
docker run hello-world
```

The software will contact the Docker daemon, pull the “hello-world” image, and create a new container based on that image.
Once it goes through all the steps, the output should inform you that ***your installation appears to be working correctly***.



Note: Try out creating containers on your own and get acquainted with some of the best practices for managing Docker containers.

Raspberry Pi Docker Images

Opposed to most other Linux distributions, Raspberry Pi is based on ARM architecture. Hence, not all Docker images will work on your Raspberry Pi.

In fact, all Docker images that are not specifically designed for ARM devices will not work. Running docker-apps designed for x86/x64 and i386 architecture on your Raspberry Pi will return an error.

Remember that when searching for images to pull from Docker Hub. Apply the **Architectures** filter to search for supported apps.

For custom solutions, the best thing to do would be to run a container based on an official image and then modify it and commit the changes to a new image.

How to Upgrade Docker on Raspberry Pi?

There is no need to re-run the convenience script to upgrade Docker. Furthermore, running the convenience script again might cause issues if it attempts to repositories that have been added already.

Upgrade Docker using the package manager with the command:

```
sudo apt-get upgrade
```

How to Uninstall Docker on Your Raspberry Pi?

Although you used a convenience script for installation, you can simply remove docker using the package manager:

```
sudo apt-get purge docker-ce
```

Note: Depending on the version of software, you may need to use an additional command to completely remove Docker: **`sudo apt-get purge docker-ce-cli`**.

To delete leftover images, containers, volumes and other related data, run the following command:

```
sudo rm -rf /var/lib/docker
```

Edited configuration files must be deleted manually.

Conclusion

You are now set to use Docker on your Raspberry Pi and develop isolated and lightweight applications using containers.

Was this article helpful? Yes No

Sofija Simic

Sofija Simic is an aspiring Technical Writer at phoenixNAP. Alongside her educational background in teaching and writing, she has had a lifelong passion for information technology. She is committed to unscrambling confusing IT concepts and streamlining intricate software installations.

Next you should read

SysAdmin, Web Servers

How to Enable SSH on Raspberry Pi {Linux, Mac OS, Windows}

February 12, 2020

This article shows you how to enable SSH on your Raspberry Pi using different methods. The guide also...

READ MORE

DevOps and Development, SysAdmin, Virtualization

How to Install Docker on CentOS 8

December 27, 2019

CentOS 8 does not provide official support for Docker. This article clearly shows you how to install a fully...

READ MORE

Virtualization, Web Servers

How to Install Docker Compose on Ubuntu 18.04

June 10, 2019

Docker Compose is a software package that modifies the behavior of Docker. Docker creates a specific...

READ MORE

Virtualization

How to Share Data Between Docker Containers

March 26, 2019

Docker allows users to run applications isolated from a host computer, without the necessity of having...

READ MORE

Live Chat Get a Quote Support | 1-855-330-1509 Sales | 1-877-588-5918

Privacy Policy GDPR Sitemap